

The generalized-Lorenz gauged A - Φ formulation

An implementation specification, transcribed from

Y.-L. Li, S. Sun, Q. I. Dai and W. C. Chew, *Finite Element Implementation of the Generalized-Lorenz Gauged A - Φ Formulation for Low-Frequency Circuit Modeling*, **IEEE Trans. Antennas Propag.** **64(10)**, 4355–4364, 2016,

with the continuum theory from

W. C. Chew, *Vector Potential Electromagnetics with Generalized Gauge for Inhomogeneous Media: Formulation*, **PIER** **149**, 69–84, 2014.

Both are in `APhi_Papers/` . `docs/GAUGE_CHOICE.md` is the decision record for *why* this gauge; this document is *what to build*.

Status: specification, nothing implemented. Equations (1)–(37) below are transcribed from the typeset paper. §7 is **my own derivation and is not in either paper** — it is flagged as such and must be checked before it is trusted.

§11 reports a spike that measured this, and it changes §7 and §8. The formulation conditions beautifully for dielectrics — flat across ten decades — and **badly for lossy conductors at low frequency**, where $\epsilon_{\text{eff}} = \epsilon - j\sigma/\omega$ creates a material contrast of $1e17$. χ must also be position-dependent, which breaks §7's symmetrization. **Read §11 before implementing anything.**

0. The convention trap — read this first

****Chew uses $e^{-i\omega t}$. We use $e^{+j\omega t}$.** His eq. (1) is $E = i\omega A - \nabla\Phi$; ours is $E = -j\omega A - \nabla\Phi$.

every i in the paper $\rightarrow -j$ here

Our convention is fixed by `src/element_matrix.cpp` ($\alpha = j\omega\sigma - \omega^2\epsilon$) and `src/problem.cpp` ($A(\cos \phi + j \sin \phi)$), and is stamped into every `.out` header.

Nothing below is a transcription error waiting to happen except this, so every equation in this document is already written in *our* convention, and anyone cross-checking against the paper must apply the mapping above rather than reading the signs off the page.

The purely real matrices in §5 are unaffected — they contain no `i`.

1. The continuum formulation, in our convention

$$\mathbf{E} = -j\omega\mathbf{A} - \nabla\Phi \quad (1)$$

$$\nabla \cdot (\epsilon \nabla \Phi) + \chi \omega^2 \Phi = -\rho \quad (2)$$

$$\nabla \times (\mathbf{v} \nabla \times \mathbf{A}) - \omega^2 \epsilon \mathbf{A} - \epsilon \nabla [(1/\chi) \nabla \cdot (\epsilon \mathbf{A})] = \mathbf{J} \quad (3)$$

$$\chi = \alpha \mu \epsilon^2 \quad (4)$$

$$(1/\chi) \nabla \cdot (\epsilon \mathbf{A}) = -j\omega\Phi \quad \text{the generalized-Lorenz gauge} \quad (5)$$

with $\mathbf{v} = 1/\mu$ as in our existing code.

(2) and (3) are completely decoupled — (2) contains no `A`, (3) contains no `Φ`. That is the whole point of this gauge: two independent systems instead of one coupled block system, with `E` recovered afterwards from (1). (2) is a scalar Helmholtz–Poisson equation; (3) is a vector Helmholtz equation carrying a gauge term.

On α . Chew's §II.A allows α to be a function of position and notes that all eigenvalues are positive for $\alpha > 0$, so (3) yields a positive-definite system. **He recommends $\chi = \mu\epsilon^2$, i.e. $\alpha = 1$ **, for the smallest condition number.

An inconsistency to resolve before coding. §II.A defines $\chi = \alpha\mu\epsilon^2$, which is position-dependent through μ and ϵ . §II.C then says *"Assuming $\chi = \mu_0\epsilon_0^2$ " and carries that constant into (21), (24) and (27). So the* *discretization as published uses a constant, free-space χ **, not the position-dependent one. Both are defensible — the gauge is a free choice — but they are different gauges and we must pick one deliberately. §8 lists this as an open decision.

Excitation

$$\nabla \cdot \mathbf{J} = -j\omega\rho \quad (13)$$

$$\nabla \cdot (\epsilon \nabla \Phi) + \chi \omega^2 \Phi = \nabla \cdot \mathbf{J} / (j\omega) \quad (14)$$

(14) is (2) with ρ eliminated through continuity, and is how the Φ -equation is driven in a current-driven problem. Chew's remark is worth keeping: **(3) resembles the current-voltage relation of an inductor and (14) that of a capacitor** — the magneto-quasi-static and electro-quasi-static halves, solved separately and to the same accuracy.

Boundary conditions

$$\Phi = \Phi_s \quad \mathbf{r} \in \Gamma_D \quad \text{Dirichlet} \quad (9)$$

$$\hat{\mathbf{n}} \cdot \epsilon \nabla \Phi = -\rho_s \quad \mathbf{r} \in \Gamma_N \quad \text{Neumann} \quad (10)$$

$$\hat{\mathbf{n}} \times \mathbf{A} = \mathbf{A}_s \quad \mathbf{r} \in \Gamma_D \quad \text{Dirichlet} \quad (11)$$

$$\hat{\mathbf{n}} \times (\mathbf{v} \nabla \times \mathbf{A}) = \mathbf{J}_s \quad \mathbf{r} \in \Gamma_N \quad \text{Neumann} \quad (12)$$

Chew notes that **with Whitney-0 for Φ and Whitney-1 for $\mathbf{A}$, the conditions on the material interfaces are satisfied automatically** — which is the property that makes a mixed conductor/dielectric port tractable, and the reason this gauge was chosen. See [GAUGE_CHOICE.md](#) §2.2.

Conductors

****Chew 2016 contains no σ .** Its conductors are PEC, imposed as boundary conditions; the volumes are dielectric. Finite conductivity enters the usual way, by folding it into a complex permittivity — in our convention

$$\epsilon_{\text{eff}} = \epsilon - j\sigma / \omega$$

so that $j\omega \epsilon_{\text{eff}} \mathbf{E} = (\sigma + j\omega\epsilon) \mathbf{E} = \mathbf{J}_{\text{total}}$. Everything below then holds with $\epsilon \rightarrow \epsilon_{\text{eff}}$. **This is not in the paper** and has one consequence worth flagging early: $\chi = \alpha\mu\epsilon^2$ becomes complex, and whether the positive-definiteness argument of §II.A survives that is not established. §8.

2. Why the gauge term cannot be discretized directly

The third term of (3) applies a divergence to $\mathbf{A}$. Discretized naively on Whitney-1 elements it **vanishes identically**, because

$$\nabla \cdot \omega_{mn} = \nabla \cdot (\lambda_m \nabla \lambda_n - \lambda_n \nabla \lambda_m) = 0$$

(15)

so a Ritz or Galerkin discretization of that term gives zero matrix entries and (3) silently degenerates to the ungauged double-curl equation.

But Whitney-1 elements are **not** solenoidal, because a constant gradient lies in their span:

$$\nabla \lambda_m = \sum_n \omega_{nm}$$

(16)

(the same fact that makes the tree-cotree null space exist at all, and the one Chew 2015 flags as the reason non-uniqueness survives curl-conforming elements).

The resolution is an intermediate quantity. Define

$$\mathbf{d} := (1/\chi) \nabla \cdot (\epsilon \mathbf{A})$$

and expand $\mathbf{d}$ in Whitney-0 elements. Chew's Fig. 2 is the justification, and it is a differential-forms argument:

$\mathbf{A}$	$\xrightarrow{* \epsilon}$	$\epsilon \mathbf{A}$	$\xrightarrow{- \nabla \cdot}$	$\nabla \cdot \epsilon \mathbf{A}$	$\xrightarrow{* \mu, * \epsilon}$	$(1/\mu \epsilon^2) \nabla \cdot \epsilon \mathbf{A}$
1-form		2-form		3-form		0-form

so $\mathbf{d}$ lands in the same space as Φ , exactly as (5) requires. Table I of the paper:

QUANTITY	SPACE	GEOMETRIC CONSTITUENT
Φ	0-form, potential space	primal nodes
$\mathbf{A}$	1-form, field space	primal edges
$\mathbf{E}$	1-form, field space	primal edges
$(1/\mu \epsilon^2) \nabla \cdot \epsilon \mathbf{A}$	0-form, potential space	primal nodes

No dual mesh is involved, which is what keeps this compatible with an ordinary primal-mesh FEM code.

3. Expansions

$$\Phi = \sum_{n=1}^{N_n} \phi_n \lambda_n(\mathbf{r}) \quad \text{Whitney-0 (P1 nodal)} \quad (17)$$

$$\mathbf{A} = \sum_{n=1}^{N_e} \mathbf{a}_n \omega_n(\mathbf{r}) \quad \text{Whitney-1 (edge)} \quad (18)$$

$$\mathbf{d} = \sum_{n=1}^{N_n} \mathbf{d}_n \lambda_n(\mathbf{r}) \quad \text{Whitney-0 (P1 nodal)} \quad (19)$$

Φ is P1 here. Ours is P2 — `src/element_matrix.cpp` builds a 10×10 Φ - Φ block over 4 vertices plus 6 edge midpoints. The de Rham/Whitney argument above is what licenses the whole construction, and it is a statement about Whitney-0. Moving Φ to P1 is the main discretization change this gauge asks for. §8.

Chew notes (20) that one could instead expand $(1/\epsilon) \nabla \cdot \epsilon \mathbf{A}$, but then the divergence theorem is needed and a surface integral term appears; he keeps (19) because it makes the third term of (3) integrable over the domain and yields a simple matrix representation.

4. Eliminating $\mathbf{d}$

Testing (19) with λ_m and using $\chi = \mu_0 \epsilon_0^2$:

$$\sum_n \langle \lambda_m, (1/\chi) \nabla \cdot (\epsilon \omega_n) \rangle \mathbf{a}_n = \sum_n \langle \lambda_m, \lambda_n \rangle \mathbf{d}_n \quad (21)$$

$$\mathbf{d} = \tilde{\mathbf{G}}^{-1} \mathbf{K}_{\text{NE}} \mathbf{a} \quad (22)$$

5. The matrices

All real; none of them carries the time convention.

$$[\tilde{G}]_{mn} = \int_{\Omega} \lambda_m \lambda_n d\Omega \quad (23)$$

$$[K_{NE}]_{mn} = \int_{\Gamma} (\epsilon/\chi) \lambda_m \omega_n \cdot \hat{n} d\Gamma - \int_{\Omega} (\epsilon/\chi) \nabla \lambda_m \cdot \omega_n d\Omega \quad (24)$$

$$[S1]_{mn} = \int_{\Omega} \nabla \lambda_m \cdot \epsilon \nabla \lambda_n d\Omega - \int_{\Gamma} \lambda_m \hat{n} \cdot \epsilon \nabla \lambda_n d\Gamma \quad (26)$$

$$[S2]_{mn} = \int_{\Omega} \chi \lambda_m \lambda_n d\Omega \quad (27)$$

$$[b_s]_m = \int_{\Omega} \lambda_m \rho d\Omega \quad (28)$$

$$[K1]_{mn} = \int_{\Omega} \nu (\nabla \times \omega_m) \cdot (\nabla \times \omega_n) d\Omega \quad (30)$$

$$[K2]_{mn} = \int_{\Omega} \epsilon \omega_m \cdot \omega_n d\Omega \quad (31)$$

$$[K_{EN}]_{mn} = \int_{\Omega} \epsilon \omega_m \cdot \nabla \lambda_n d\Omega \quad (32)$$

$$[b_k]_m = \int_{\Omega} \omega_m \cdot J d\Omega \quad (33)$$

The two systems

$$\Phi\text{-equation:} \quad (S1 - \omega^2 S2) \phi = b_s \quad (25)$$

$$A\text{-equation:} \quad (K1 - \omega^2 K2 - K_{EN} \tilde{G}^{-1} K_{NE}) a = b_k \quad (29)$$

Dimensions

$S1, S2, \tilde{G}$	$N_n \times N_n$
$K1, K2$	$N_e \times N_e$
K_{EN}	$N_e \times N_n$
K_{NE}	$N_n \times N_e$
ϕ, b_s	$N_n \times 1$
a, b_k	$N_e \times 1$

For a large 3-D mesh $N_n/N_e \approx 1/7.3$, so the A-system costs about **13.7 %** more unknowns than an E formulation. Chew measures 15.04 % and 15.35 % on his two examples.

Static bounds, for preconditioning and for sanity

$$\begin{aligned} K &= K1 - \omega^2 K2 - K_{EN} \bar{G}^{-1} K_{NE} & K' &= K1 - K_{EN} \bar{G}^{-1} K_{NE} & (35), (37) \\ S &= S1 - \omega^2 S2 & S' &= S1 & (34), (36) \end{aligned}$$

K' and S' are the $\omega = 0$ limits and are **frequency-independent**. Chew's reported condition numbers: K stays bounded by K' at $5.94e5$ and $1.21e6$ across his two cases and the whole frequency range, against the E formulation's $1.04e17$ at 10 kHz.

6. Solving — direct is explicitly supported

Worth recording plainly, because the first reading of this paper suggested otherwise. §II.D: *sparse direct solvers such as UMFPACK and CHOLMOD are suitable candidates* for problems of tens of thousands of unknowns, and §III states that **UMFPACK multifrontal LU is applied first** to make the solution as accurate as possible. The ILU/GMRES material is an optimization for wideband sweeps — the frequency-independent K' can be factored once and reused at every frequency — **not a requirement of the method**.

So this composes with our MUMPS direct path rather than fighting it.

7. Avoiding the sparse approximate inverse — our variant

This section is my derivation. It is not in either paper and has not been tested. It must be checked symbolically and numerically before it is relied on. Everything above is transcribed; everything here is not.

$\bar{G}^{-1}$ is the problem. $\bar{G}$ is the nodal mass matrix, so $\bar{G}^{-1}$ is dense, and $K_{EN} \bar{G}^{-1} K_{NE}$ formed explicitly would destroy sparsity. Chew's answer is a **sparse approximate inverse** — on his examples $\bar{G}^{-1}$ is replaced by an SAI with 9057 and 7147 nonzeros, 98.63 % and 99.98 % sparse. That is an approximation adopted to keep the system $N_e \times N_e$ for iterative solving.

We factor directly, so we can avoid it entirely by keeping $\mathbf{d}$ as unknowns. From (22), $\tilde{\mathbf{G}} \mathbf{d} = \mathbf{K}_{NE} \mathbf{a}$, and (29) becomes $(\mathbf{K}_1 - \omega^2 \mathbf{K}_2) \mathbf{a} - \mathbf{K}_{EN} \mathbf{d} = \mathbf{b}_k$:

$$\begin{bmatrix} \mathbf{K}_1 - \omega^2 \mathbf{K}_2 & -\mathbf{K}_{EN} \\ \mathbf{K}_{NE} & -\tilde{\mathbf{G}} \end{bmatrix} \begin{bmatrix} \mathbf{a} \\ \mathbf{d} \end{bmatrix} = \begin{bmatrix} \mathbf{b}_k \\ \mathbf{0} \end{bmatrix}$$

Everything in it is sparse and exact. The cost is $+Nn$ unknowns — about 14 % on top of the A-system, i.e. roughly the same again as the formulation already costs over an E formulation.

It can be made symmetric. Ignoring the surface term, (24) is

$$[\mathbf{K}_{NE}]_{mn} = -(1/\chi) \int_{\Omega} \epsilon \nabla \lambda_m \cdot \omega_n \, d\Omega = -(1/\chi) [\mathbf{K}_{EN}]_{nm}$$

so $\mathbf{K}_{NE} = -(1/\chi) \mathbf{K}_{EN}^T$. Scaling the second block row by χ :

$$\begin{bmatrix} \mathbf{K}_1 - \omega^2 \mathbf{K}_2 & -\mathbf{K}_{EN} \\ -\mathbf{K}_{EN}^T & -\chi \tilde{\mathbf{G}} \end{bmatrix} \begin{bmatrix} \mathbf{a} \\ \mathbf{d} \end{bmatrix} = \begin{bmatrix} \mathbf{b}_k \\ \mathbf{0} \end{bmatrix} \quad \text{symmetric}$$

which is what MUMPS's complex-symmetric path wants.

Three things to check before trusting this:

1. **The surface term of (24)** — $\int_{\Gamma} (\epsilon/\chi) \lambda_m \omega_n \cdot \hat{n} \, d\Gamma$ — is dropped above. It must either vanish under our boundary conditions or be carried, and if carried it breaks the exact symmetry.
2. **** χ inside the scaling**** must be constant for the row scaling to be a congruence. If we adopt a position-dependent χ (§1), this derivation needs redoing.
3. **Conditioning.** The block system is larger and has a zero-ish structure in the $(2,2)$ corner; whether its condition number behaves as well as Chew's reported $\mathbf{K}$ is an empirical question, and is the first thing the spike should measure.

If any of these fails, Chew's SAI route remains available and is the fallback.

8. Open decisions

** χ constant or position-dependent? **	§II.A defines $\chi = \alpha\mu\epsilon^2$; §II.C discretizes with $\chi = \mu_0\epsilon_0^2$ constant. Constant makes §7's symmetrization a clean congruence. Position-dependent is the "generalized" gauge as stated. Pick deliberately and record why.
$\Phi: P2 \rightarrow P1?$	The Whitney/de Rham argument is about Whitney-0. Our Φ is P2. Either move Φ to P1 or redo the compatibility analysis for P2 — the latter is research, not implementation.
Lossy conductors	Fold σ into $\epsilon_{\text{eff}} = \epsilon - j\sigma/\omega$. Not in the paper. Makes χ complex; the positive-definiteness argument of §II.A is then unproven.
** $\bar{G}^{-1}$: block system or SAI? **	§7 versus Chew's SAI. Decide on measured conditioning, not on preference.
Per-region boundary conditions on a port face	Needed for the mixed port, which is the acceptance test (§10). The formulation supplies it through the Γ_D / Γ_N partition of (9)–(12); our parser rejects per-surface BCs today. Parser and binding work, independent of the gauge, and can proceed in parallel.

9. What maps onto existing code

From `include/aphi_solver/element_matrix.hpp`, we already build per-tet:

OURS	THEIRS	STATUS
A-A curl part, $\mathbf{v} \int (\nabla \times \mathbf{W}_i) \cdot (\nabla \times \mathbf{W}_j)$	** $K1$ ** (30)	identical
A-A mass part, $\alpha \int \mathbf{W}_i \cdot \mathbf{W}_j$, $\alpha = j\omega\sigma - \omega^2\epsilon$	** $-\omega^2 K2$ ** (31)	same shape; our α already carries σ via the complex- ϵ route
A- Φ coupling, $\beta \int \mathbf{W}_i \cdot \nabla \mathbf{S}_b$, $\beta = \sigma + j\omega\epsilon$	** K_{EN} ** (32)	same shape, different coefficient and Φ space
Φ - Φ , $\beta \int \nabla \mathbf{S}_a \cdot \nabla \mathbf{S}_b$	** $S1$ ** (26)	same shape, coefficient ϵ not β

New work:

- ** $\bar{G}$ ** (23) — nodal mass. Trivial; the quadrature exists.

- **S2** (27) — nodal mass weighted by χ . Trivial given $\bar{G}$.
- **K_NE** (24) — the genuinely new one. A volume term that is K_{EN}^T up to the ϵ/χ weighting, plus a **surface** term over Γ . Our assembly has no surface-integral path for this block today.
- **The gauge term's coupling** — either the block system of §7 or an SAI.
- **Decoupling the solve** — (25) and (29) are independent, so the driver changes from one factorization to two.
- **Removing tree–cotree** from the A path.

What goes away: `tree_cotree.cpp`, `gauge_variants.cpp` and the gauging in `dof_map.cpp` cease to be load-bearing. Keep `07_GaugeInvariance` — it becomes the test that Φ *no longer* moves.

10. Validation plan

The acceptance test is the mixed-material port

This is why the gauge is being replaced at all. `ROADMAP.md` Phase 03 names a port face with conductor on one side and dielectric on the other, inside the domain, as the reason tree–cotree is inadequate; it is corroborated by the project owner's experience of a commercial solver running such a port to completion and returning wrong answers (`GAUGE_CHOICE.md` §1.3). Everything else in this list is a necessary condition. **This one is the goal**, and the gauge has not succeeded until it passes.

The formulation already provides what a mixed port needs, which the first draft of this section under-stated. Eqs (9)–(12) partition the boundary into Γ_D and Γ_N **separately for Φ and for A** . A mixed port face is exactly that partition:

PART OF THE FACE	Φ	A
conductor	$\Phi = \Phi_S$ — Dirichlet (9)	$\hat{n} \times A = A_S$ — Dirichlet (11)
dielectric	$\hat{n} \cdot \epsilon \nabla \Phi = -\rho_S$ — Neumann (10)	$\hat{n} \times (\nu \nabla \times A) = J_S$ — Neumann (12)

and Chew's remark that with Whitney-0 Φ and Whitney-1 A the **interface conditions are satisfied automatically** is what makes the conductor/dielectric boundary inside the face unproblematic.

The gap is on our side and it is bounded. `src/input_file.cpp` currently fails with "*per-surface boundary conditions are not supported yet*", and every port type we have imposes a single potential over the whole tagged surface. So a mixed port needs **per-region boundary conditions on a port face** — Dirichlet on the conductor part, natural on the dielectric part. That is parser and binding work, not a reformulation, and it is independent enough to proceed in parallel with the gauge.

The baseline cannot be built yet, and that sets the order

A mixed port is not expressible by this solver today. `[boundary]` "takes no name and applies to the whole outer boundary" — per-surface conditions are explicitly unimplemented — and every port type imposes one potential over the whole tagged surface. So building the coax now would produce one of two useless results:

- tag the whole cross-section as one port → the dielectric is shorted to the conductor's potential → the ill-posed garbage the first case 08 attempt produced;
- tag only the conductors → it is not a mixed port, and case 05 has already measured what an interior conductor cut does.

Either way it would measure the port model's inability to pose the problem, not the gauge's failure to solve it — the same confound that wrecked the first attempt, on a larger mesh. "Measure the failure before fixing it" does not rescue it: a failure cannot be measured on a problem that cannot be stated.

So the dependency is strict rather than parallel:

per-region port BCs -> coax baseline -> mixed-port acceptance test

The first case 08 attempt is worth remembering here. A straight rod with an internal cut has no return path, so the system was ill-posed; both the mixed face and the conductor-only control came back wrecked, and it was the *control* that showed the geometry was at

fault rather than the gauge. Any rebuild needs a closed circuit — coax, up the centre and back through the shield.

In order

Steps 1 and 2 are genuinely independent and can run in parallel. Nothing after them can start early.

1. **Spike the gauge term** — assemble `K_NE` and `G` on `07_GaugeInvariance`'s 8,365-tet mesh. Is the §7 block system complex-symmetric, does MUMPS factor it, and how does its condition number compare with the tree-gauged system? **No new geometry, no port work, commits to nothing, and it retires the largest unknown in the plan.**
2. **Per-region boundary conditions on a port face** — Dirichlet on the conductor part, natural on the dielectric part, shaped by the `r_D / r_N` partition of (9)–(12). Parser and binding work, not reformulation. **On the critical path regardless of the gauge**, because without it there is no acceptance test.
3. **Case 08 built and failing** — the coax mixed port, demonstrated broken under tree–cotree, with numbers. Nearly free once step 2 exists.
4. **Reproduce Chew's Fig. 5** qualitatively — condition number flat against frequency where the E formulation blows up. His `5.94e5` and `1.21e6` are the target order.
5. **** 07_GaugeInvariance inverted**** — Φ must now be mesh-independent. The permutation harness stays; the *potential-moves* assertions flip.
6. **Case 05's local field** — the 8 % tree-dependence of `J` within 2 mm of the cut (`GAUGE_CHOICE.md` §1.3) must go away. A necessary condition, and the only one measurable on an existing case today.
7. **Cases 01–06 unchanged** on every field quantity. This is a reformulation, not a change of physics.
8. **The mixed port passes** — step 3's baseline, now correct. **Tree-independence is not the test here**: once the tree is gone there is no tree to depend on, so the standard is that the local field on and near the port face **converges under mesh refinement**.

11. Spike results — measured, and they change §7 and §8

`tools/gauge_spike/` assembles §5's matrices on 518 tets and computes exact condition numbers by dense SVD. It ran before any C++ was written, which is the only reason the findings below cost a day rather than a month.

The formulation does what Chew claims — for dielectrics

His own materials: dielectric everywhere, no σ .

FREQ (HZ)	COND GAUGED	COND UNGAUGED
1e0	2.492e+02	1.884e+18
1e4	2.492e+02	2.584e+14
1e10	3.907e+02	5.729e+02

Flat across ten decades while the ungauged system degrades by sixteen orders as $\omega \rightarrow 0$. That is Fig. 5 of the paper reproduced in character, and it is what validates the assembly — every other number here depends on it.

It does not survive σ folded into ϵ

FREQ (HZ)	COND GAUGED	COND UNGAUGED	E CONTRAST
1e0	2.046e+22	1.655e+06	2.317e+17
1e4	9.272e+17	5.745e+02	2.317e+13
1e10	1.527e+10	1.118e+09	2.317e+07

The condition number **tracks the ϵ contrast** at about `contrast × 1e4`, and below roughly 1 GHz **the gauged system is worse than the ungauged one.**

Not a bug: $\epsilon_{\text{eff}} = \epsilon - j\sigma/\omega$ gives $\sigma/\omega\epsilon = 2.3e17$ between copper and dielectric at 1 Hz. **Chew has no finite conductivity** — PEC boundary conditions, dielectric volumes — so his ϵ varies by 4.5. ϵ_{eff} is §1's addition, and it is what breaks the conditioning.

This qualifies the roadmap's low-frequency rationale, sharply. Phase 03 wants this gauge partly to remove the low-frequency floor. For a *dielectric* problem it does, completely. For **lossy metal at low frequency it makes conditioning worse**, monotonically in $\sigma/\omega\epsilon$.

It does not invalidate the choice — we want this gauge for the **embedded interface conditions** at a mixed port (§10), which the above does not touch. But "fixes low-frequency breakdown" must not be claimed for conductors without qualification.

At THz the problem recedes. The contrast falls as $1/\omega$: $2.3e7$ at 10 GHz and about $2.3e5$ at 1 THz, where conditioning should be unremarkable. The difficulty is specific to the low-frequency end with good conductors.

χ must be position-dependent — and that breaks §7's symmetry

COND AT 1 HZ

$\chi = \mu_0 \epsilon_0^2$ constant — his §II.C	7.026e+37
$\chi = \mu \epsilon^2$ per element — his §II.A	2.046e+22

Fifteen orders, because $\epsilon^2/\chi = 1/\mu = v$ self-normalizes to the curl–curl coefficient
 only when χ carries the local ϵ . §8's open question is therefore answered: position-dependent.

But §7's symmetrization is a global row scaling by χ , a congruence only for constant χ . Measured per element, K_{NE} is no longer a scalar multiple of K_{EN}^T — residual **O(1)**, not $1e-16$.

As derived, the block system is symmetric or well-conditioned, not both. The options are a per-element scaling that is still a congruence, an unsymmetric system, or Chew's SAI route.

What §7 got right, and what it got wrong

Right: with constant χ the block system is exactly complex-symmetric — measured $1e-21$ to $8e-17$. The derivation $K_{NE} = -(1/\chi)K_{EN}^T$ holds.

Wrong: §7 ignored scaling. $\chi \approx 1e-28$ puts the $(2,2)$ block $1e40$ below the $(1,1)$ block, and the first spike run reported $1e46$ because of it. The scaling c is free — it cancels in the Schur complement — so it can be chosen for conditioning, and $c^2 = \|K_{uu}\| / (\chi \|G\|)$ recovered 26 orders. **Any block form must be scaled deliberately;** the naive one is unusable.

Why the controls are the whole point

The first two spike runs concluded the formulation was unusable, at $1e38$. Both were wrong — the first was my block scaling, the second was finite conductivity, which is not in the paper. Only running **Chew's materials unchanged** showed the assembly had been right the whole time.

Without that control this document would now be recommending against a sound method.

Revised open questions

**How to handle σ **	ϵ_{eff} is measured to wreck conditioning below ~ 1 GHz. To evaluate: keep σ explicit rather than folded into ϵ ; a different χ inside conductors; or accept it, given THz is the target. This is now the main obstacle, and nothing should be implemented before it is settled.
**Symmetry with per-element χ **	As above. Symmetric or well-conditioned, not both, as currently derived.
The surface term of (24)	Still omitted in the spike. Must vanish under our boundary conditions or be carried; carrying it may break symmetry independently of χ .